HACKSAW BLADE WITH FRAME

1. Description

A **hacksaw blade with frame** is a hand tool used for **cutting metal, plastic, and thin sections of material**. It consists of two main parts:

- **Frame**: Holds the blade under tension (fixed or adjustable type).
- **Blade**: A fine-toothed strip made of **high carbon steel (HCS)** or **bi-metal (HSS edge + flexible backing)**.

The blade is mounted with teeth facing forward so that cutting occurs during the forward stroke. Frames may include a **tensioning screw** to keep the blade tight and straight during operation.

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2. How to Operate

Procedure:

1. Select a suitable blade (based on TPI and material).
2. Fix the blade into the frame with **teeth pointing forward**.
3. Tighten the frame using the wing nut/tension screw.
4. Mark the cutting line on the workpiece.
5. Clamp the workpiece securely in a vice.
6. Hold the hacksaw properly (dominant hand on handle, other on frame).
7. Begin cutting with smooth, controlled strokes.
8. Apply pressure only on the forward stroke.
9. Maintain straight alignment throughout cutting.
10. Reduce pressure at the end to prevent blade breakage.

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3. Do’s and Don’ts

Do’s

- Use correct blade pitch (TPI) for material thickness.
- Keep the blade properly tensioned.
- Use full-length strokes for efficiency.
- Maintain steady cutting speed.
- Use cutting fluid when needed.

Don’ts

- Do not apply force on return stroke.
- Do not use a loose or bent blade.
- Do not twist the frame while cutting.
- Do not cut without clamping the workpiece.
- Do not rush or apply excessive pressure.

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4. Safety Measures

- Wear **safety goggles** to avoid eye injury.
- Ensure firm clamping of the workpiece.
- Keep hands clear of the blade path.
- Maintain proper posture while cutting.
- Replace damaged or worn blades immediately.
- Work in a clean and well-lit environment.

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5. Real Image (Reference)

Handle Hacksaw

Mini Frame Hacksaw

Long Frame Hacksaw